

PROJECT NAME : 2D RUNNING GAME

Project Requirement Analysis

Project Objective:

The objective of this project is to develop a simple 2D Running Game using Java Swing. The player controls a character that runs continuously and avoids obstacles. The game ends when the player collides with an obstacle, and the final score is displayed.

Functional Requirements:

1. Character Movement:

The player character will move continuously, and the user can control its movement to avoid obstacles.

2. Obstacle Generation:

Obstacles will appear randomly on the screen and move toward the player.

3. Collision Detection:

The system will detect collisions between the player and obstacles. If a collision occurs, the game will end.

4. Score Calculation:

The score will increase based on the player's survival time during gameplay.

5. Game Over Screen:

A game over message and the final score will be displayed when the player loses.

6. Play Again Button:

This button will allow the player to restart the game without closing the application.

7. Exit Button:

This button will close the game application when selected.

Tools and Technologies:

1. Java:

Java is the main programming language used to develop the game. It is used to implement game logic, player movement, obstacle generation, collision detection, and score calculation.

2. Java Swing:

Java Swing is used to create the graphical user interface (GUI) of the game, including the game window, buttons, panels, and score display.

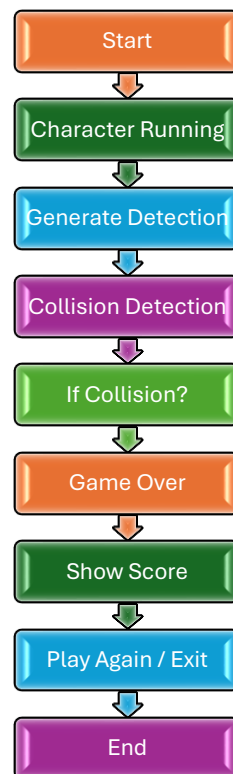
3. GitHub:

GitHub is used for version control and project management. It helps track weekly progress and store project files online.

4. IntelliJ IDEA / NetBeans:

IntelliJ IDEA or NetBeans is used as the Integrated Development Environment (IDE) for writing, testing, debugging, and managing the project source code.

Workflow Diagram :



Development Environment Setup:

- Java Development Kit (JDK) is already installed and configured in the system.
- An Integrated Development Environment (IDE) such as IntelliJ IDEA / NetBeans is installed and used for development.
- The environment is properly configured for compiling, running, and debugging Java Swing applications.